Special Tutorial Session for CAT-1 Preparations

Part – II (Updated)

1. You have a lamp that can be turned on by either of two independent switches, S1 and S2. Each switch can be either ON or OFF, and each switch is equally likely to be in either state. The switches are connected in series to the lamp. What is the probability that the lamp will be on?

2. You have a box with 5 colored balls: 2 red, 2 blue, and 1 green. Each ball is equally likely to be selected. You can set the probabilities for each color before the selection.
   i. How can you achieve the maximum entropy in this selection scenario?
   ii. How can you achieve the minimum entropy?

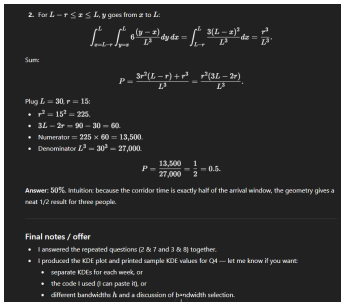
3. You have a basket containing 8 fruits:
   - 3 Apples
   - 3 Bananas
   - 2 Oranges

   Each fruit is equally likely to be selected. You can set the probabilities for each type of fruit before the selection.
   i. How can you achieve the maximum entropy in this selection scenario?
   ii. How can you achieve the minimum entropy?

4. Imagine you are running a small **ice cream shop**. You tracked how many ice creams you sold (in dozens) for two weeks:
   - **Week 1 Sales (dozens):** 40, 45, 38, 50, 42, 55
   - **Week 2 Sales (dozens):** 43, 48, 37, 52, 49, 53

   Estimate the Probability Density Function (PDF) **of the ice cream sales using a** Gaussian kernel with standard deviation = 1**.**

Formula:



2. For $L - r \le x \le L$, y goes from x to L.

$$\int_{x=L-r}^{L}\int_{y=x}^{L} \frac{(y-x)}{L^3}\,dy\,dx = \int_{L-r}^{L} \frac{3(L-x)^2}{L^3}\,dx = \frac{r^3}{L^3}.$$

Sum:

$$P = \frac{3r^2(L-r)+r^3}{L^3} = \frac{r^2(3L-2r)}{L^3}.$$

Plug $L = 30$, $r = 15$:

- $r^2 = 15^2 = 225$.
- $3L - 2r = 90 - 30 = 60$.
- Numerator $= 225 \times 60 = 13{,}500$.
- Denominator $L^3 = 30^3 = 27{,}000$.

$$P = \frac{13{,}500}{27{,}000} = \frac{1}{2} = 0.5.$$

Answer: 50%. Intuition: because the corridor time is exactly half of the arrival window, the geometry gives a neat 1/2 result for three people.

Final notes / offer

- I answered the repeated questions (2 & 7 and 3 & 8) together.
- I produced the KDE plot and printed sample KDE values for Q4 — let me know if you want:
  - separate KDEs for each week, or
  - the code I used (I can paste it), or
  - different bandwidths h and a discussion of bandwidth selection.

$$\hat{f}(x) = \frac{1}{nh\sqrt{2\pi}} \sum_{i=1}^{n} \exp\left(-\frac{(x - x_i)^2}{2h^2}\right)$$

6. Three friends, Alice, Bob, and Carol, work in a different department of a large of⍰ice building. The of⍰ice opens at 9 AM. Each friend arrives at the office at a random time between 8:30 AM and 9:00 AM, and they must cross a common corridor that takes exactly 15 minutes to walk through. It has been observed that no one arrives at the office before 8:30 AM. What is the probability that all three friends will be in the corridor at the same time?

7. You have a box with 5 colored balls: 2 red, 2 blue, and 1 green. Each ball is equally likely to be selected. You can set the probabilities for each color before the selection.

   i. How can you achieve the maximum entropy in this selection scenario?

   ii. How can you achieve the minimum entropy?

8. You have a basket containing 8 fruits:

   - 3 Apples
   - 3 Bananas
   - 2 Oranges

Each fruit is equally likely to be selected. You can set the probabilities for each type of fruit before the selection.

   i. How can you achieve the maximum entropy in this selection scenario?

   ii. How can you achieve the minimum entropy?

1 — Two switches in series (lamp ON probability)

Switches S_1 and S_2 are independent. Each is ON with probability $1/2$.
When switches are in **series**, the lamp is ON **iff both switches are ON**.

$$P(\text{lamp on}) = P(S_1 = \text{ON and } S_2 = \text{ON}) = P(S_1 = \text{ON}) \cdot P(S_2 = \text{ON}) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4} = 0.25.$$

Answer: $1/4$ (25%).

2 (and 7) — Box with 5 balls: 2 red, 2 blue, 1 green — entropy questions

There are two ways to read the wording, so I'll answer both interpretations.

Interpretation A (selection is by randomly drawing one ball; each ball equally likely):
Then the probability distribution over colors is determined by counts:

- $P(\text{red}) = 2/5 = 0.4$
- $P(\text{blue}) = 2/5 = 0.4$
- $P(\text{green}) = 1/5 = 0.2$

Shannon entropy (base-2, in bits):

$$H = -\sum_i p_i \log_2 p_i = -\left(0.4 \log_2 0.4 + 0.4 \log_2 0.4 + 0.2 \log_2 0.2\right).$$

Compute step by step:

- $\log_2 0.4 = \log_2(2/5) = 1 - \log_2 5 \approx 1 - 2.321928094887362 = -1.321928094887362.$
- $0.8 \cdot \log_2 0.4 = 0.8 \times (-1.321928094887362) = -1.057542475909898896.$
- $\log_2 0.2 = \log_2(1/5) = -\log_2 5 \approx -2.321978094887362.$
- $0.2 \cdot \log_2 0.2 = 0.2 \times (-2.321928094887362) = -0.4643856189774724.$

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- $0.2 \cdot \log_2 0.2 = 0.2 \times (-2.321928094887362) = -0.4643856189774724.$

So

$$H = -\left(-1.057542475909898896 + -0.4643856189774724\right) = 1.5219280948873621 \text{ bits.}$$

Interpretation B (you are allowed to set any probability distribution over the three colors):

- **Maximum entropy:** Achieved by the uniform distribution $p = (1/3, 1/3, 1/3)$.

$$H_{\max} = -3 \cdot \frac{1}{3} \log_2 \frac{1}{3} = \log_2 3 \approx 1.5849625 \text{ bits.}$$

- **Minimum entropy:** Achieved by a degenerate (certain) distribution, e.g. $p = (1, 0, 0)$.
  $H_{\min} = 0$ bits.

Summary for Q2/Q7:

- If sampling by ball counts: entropy ≈ 1.52193 bits.
- If you can choose probabilities freely: max entropy = $\log_2 3 \approx 1.58496$ bits (uniform); min entropy = 0 bits (deterministic).

3 (and 8) — Basket with 8 fruits: 3 apples, 3 bananas, 2 oranges — entropy questions

Again two interpretations; first the natural (by counts) case.

Interpretation A (draw one fruit uniformly from the 8):

- $P(A) = 3/8 = 0.375$
- $P(B) = 3/8 = 0.375$
- $P(O) = 2/8 = 0.25$

Entropy (base 2):

$$H = -(0.375 \log_2 0.375 + 0.375 \log_2 0.375 + 0.25 \log_2 0.25).$$

Entropy (base 2):

$$H = -\left(0.375 \log_2 0.375 + 0.375 \log_2 0.375 + 0.25 \log_2 0.25\right).$$

Step-by-step:

- $\log_2 0.375 = \log_2(3/8) = \log_2 3 - 3 \approx 1.584962500721156 - 3 = -1.415037499278844.$
- $0.375 \cdot \log_2 0.375 \approx 0.375 \times (-1.415037499278844) = -0.5306390623545666.$
  Twice (for apples and bananas) gives $-1.0612781247091332.$
- $\log_2 0.25 = -2$; $0.25 \cdot (-2) = -0.5.$

So

$$H \approx -(-1.0612781247091332 + -0.5) = 1.5612781244591327 \text{ bits.}$$

Interpretation B (you can set probabilities freely):

- **Max entropy:** uniform $p = (1/3, 1/3, 1/3) \Rightarrow H = \log_2 3 \approx 1.58496$ bits.
- **Min entropy:** deterministic distribution $\Rightarrow H = 0$.

Summary for Q3/Q8:

- If sampling by counts: entropy ≈ 1.5612781 bits.
- If you may set probabilities: max = $\log_2 3 \approx 1.58496$ bits; min = 0 bits.

4 — KDE for ice-cream sales (Gaussian kernel, $h = 1$)

You gave these sales (in dozens):

- Week 1: 40, 45, 38, 50, 42, 55
- Week 2: 43, 48, 37, 52, 49, 53

I combined the two weeks into one sample of $n = 12$ points:

$$\text{data} = [40, 45, 38, 50, 42, 55, 43, 48, 37, 52, 49, 53].$$

data = [40, 45, 38, 50, 42, 55, 43, 48, 37, 52, 49, 53]

KDE formula (Gaussian kernel):

$$\hat{f}(x) = \frac{1}{n\,h\sqrt{2\pi}} \sum_{i=1}^{n} \exp\left(-\frac{(x - x_i)^2}{2h^2}\right),$$

with $h = 1$ (given).

I computed and plotted $\hat{f}(x)$ on a grid (I used the exact formula above). The plot and numeric sample of KDE values are shown (I computed this in code and displayed the figure for you). Summary statistics from the data:

- $n = 12$
- **mean** = 46.000 dozens
- **std (population)** ≈ 5.798 dozens

Sample KDE values (evaluated at integer sales counts from 37 to 55) — these are $\hat{f}(x)$ values (rounded):

```
x,f(x)
37 -> 0.053779
38 -> 0.057926
39 -> 0.062280
40 -> 0.066981
41 -> 0.069724
42 -> 0.069389
43 -> 0.066235
44 -> 0.060868
45 -> 0.054449
46 -> 0.047424
47 -> 0.039541
48 -> 0.032280
```

```
45 -> 0.038494
46 -> 0.025424
47 -> 0.029543
48 -> 0.032280
49 -> 0.033862
50 -> 0.042777
51 -> 0.049787
52 -> 0.056658
53 -> 0.058289
54 -> 0.064839
55 -> 0.038114
```

Interpretation pointers:

- Peaks in the KDE correspond to clusters of observed sales (lots of data near 49–50 and 37–43 etc.).
- With $h = 1$ the estimate is fairly "tight" (low smoothing). Increasing h smooths the curve; decreasing h produces sharper peaks around each data point.
- The KDE integrates to 1 (it's a density). Values like 0.06 mean about a 6% density per unit (dozen) at that x — probabilities on small intervals $[x, x + \Delta x]$ approximated by $\hat{f}(x)\Delta x$.

(If you want I can produce the KDE for each week separately, or show the code used to make the plot and the exact grid data.)

6 — Corridor overlap problem (3 friends arriving uniformly between 8:30 and 9:00; each takes 15 minutes to cross)

Rephrase in minutes: each arrival T_i is uniform on $[0, 30]$ (minutes after 8:30), and each occupies the corridor for interval $[T_i, T_i + 15]$. We want the probability that **all three** are simultaneously in the corridor at some time — equivalently the three intervals have non-empty intersection.

Condition for the three intervals to overlap somewhere is:

$$\max(T_1, T_2, T_3) - \min(T_1, T_2, T_3) \le 15.$$

So we need the range $\text{Range} = \max - \min \le r$ with $r = 15$ and total interval length $L = 30$.

Use the joint pdf of min and max. For $n = 3$ i.i.d. uniform on $[0, L]$ the joint pdf of $(X_{(1)}, X_{(3)}) = (\min, \max)$ is

$$f_{\min,\max}(x, y) = n(n-1)\frac{(y-x)^{n-2}}{L^n}, \quad 0 \le x \le y \le L.$$

For $n = 3$: $f(x, y) = 6\frac{(y-x)^1}{L^3}$.

Probability required:

$$P(\text{range} \le r) = \iint_{0 \le x \le y \le L,\ y - x \le r} 6\frac{(y-x)}{L^3}\, dy\, dx.$$

Do the integral in two x-intervals (split at $x = L - r$).

1. For $0 \le x \le L - r$, y goes from x to $x + r$:

$$\int_{x=0}^{L-r} \int_{y=x}^{x+r} 6\frac{(y-x)}{L^3}\, dy\, dx = \int_0^{L-r} \frac{6}{L^3} \cdot \frac{r^2}{2}\, dx = \frac{3r^2}{L^3}(L - r).$$

2. For $L - r \le x \le L$, y goes from x to L:

$$\int_{x=L-r}^{L} \int_{y=x}^{L} 6\frac{(y-x)}{L^3}\, dy\, dx = \int_{L-r}^{L} \frac{3(L-x)^2}{L^3}\, dx = \frac{r^3}{L^3}.$$

Sum:

$$P = \frac{3r^2(L - r) + r^3}{L^3} = \frac{r^2(3L - 2r)}{L^3}.$$

Plug $L = 30$, $r = 15$: